Zaměň pořadí integrace

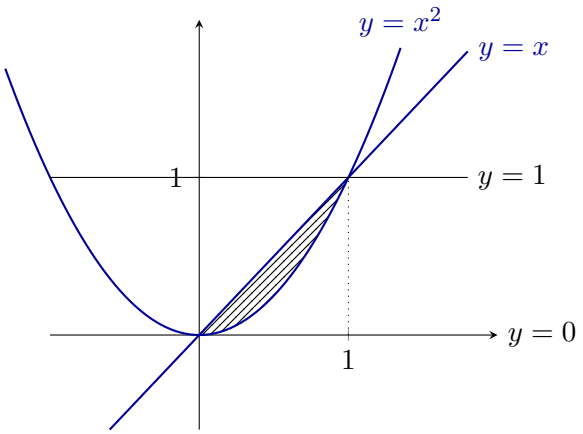
$$\int_0^1 \int_y^{\sqrt{y}} f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_{y=0}^{y=1} \left(\int_{x=y}^{x=\sqrt{y}} f(x,y)\,\mathrm{d}x \right) \mathrm{d}y$$

Pás pro y:	$\Rightarrow$	Pás pro x:
$y = 1$		$x = 0$
$y = 0$		$x = 1$

Meze pro x v opačném pořadí:

- $x = y \longrightarrow \boxed{y = x}$
- $x = \sqrt{y}\ /^2, \quad x^2 = y \longrightarrow \boxed{y = x^2}$

1. obrázek



Průsečík $f = g$:

$$\begin{aligned} x &= x^2 \\ x^2 - x &= 0 \\ x(x-1) &= 0 \\ x = 0, &\quad x = 1 \end{aligned}$$

$$\boxed{\begin{array}{c} 0 \le x \le 1 \\ x^2 \le y \le x \end{array}} \quad \Longrightarrow \quad \underbrace{\boxed{\int_0^1 \left(\int_{x^2}^{x} f(x,y)\,\mathrm{d}y \right) \mathrm{d}x}}_{\text{VÝSLEDEK}}$$